TryHackMe — Pre-Security Learning Path

What is Networking?

Learning Notes — Networks, IP, MAC & Ping

"Networks are the foundation of all communication — and the foundation of all attacks."

Networks & Internet	IP Addresses	MAC Addresses	Ping & ICMP

1. What Is a Network?

A network is any collection of connected devices that can communicate with each other. Networks come in all shapes and sizes — from two computers in a bedroom to billions of devices spanning the entire globe. Understanding networks is the single most important foundation for a career in cyber security, because every attack and every defence travels through one.

Why Networks Matter in Cyber Security

Every cyber attack — whether it is a phishing email, a SQL injection, or ransomware spreading across an organisation — travels over a network. To attack or defend systems effectively, you must first understand how devices communicate, how they are identified, and how data moves between them.

1.1 The Human Analogy

TryHackMe uses a helpful real-world analogy: think of a network like a friendship circle. You are all connected because of shared interests, hobbies, and experiences. Networks of devices are connected in the same way — through shared infrastructure, protocols, and addressing systems that allow them to communicate.

- Small networks: two computers connected at home to share files or a printer.

- Medium networks: an office LAN connecting dozens of workstations, servers, and printers.
- Large networks: multinational corporations with thousands of devices across continents.
- The Internet: the largest network of all — a global network of networks.

2. What Is the Internet?

The Internet is not a single network — it is a vast collection of many smaller networks all interconnected together. The key significance of the Internet is that it allows entirely different and separate networks to communicate with each other seamlessly.

2.1 A Brief History

Year	Milestone
1960s	ARPANET — the first network, developed by the US Department of Defense. Designed to survive a nuclear attack by rerouting data around damaged nodes.
1970s–80s	TCP/IP protocols developed, giving all networks a common language for communication.
1989	Tim Berners-Lee invents the World Wide Web (WWW) — making the Internet accessible to ordinary people via browsers and web pages.
1990s	Commercial Internet opens up; rapid growth of websites, email, and online services.
Today	Billions of devices connected — smartphones, laptops, IoT sensors, servers, and more — all communicating over the same underlying network infrastructure.

Key Fact

The World Wide Web (WWW) and the Internet are NOT the same thing. The Internet is the underlying infrastructure (cables, routers, protocols). The Web is a service that runs on top of it — web pages accessed via HTTP/HTTPS. Email, video calls, and gaming also run on the Internet but are not the Web.

3. Identifying Devices on a Network

For two devices to communicate over a network, they must be able to identify each other. Every device connected to a network has two key identifiers: an IP address (logical identifier) and a MAC address (physical identifier). Understanding both is fundamental to networking and security.

3.1 IP Addresses (Internet Protocol)

IP Address — The Logical Identifier

An IP address is a set of numbers assigned to a device that identifies it on a network. It works like a postal address — it tells the network where to deliver data. IP addresses can change over time (they can be reassigned to different devices) but cannot be active on more than one device simultaneously within the same network.

- An IP address is divided into four sections called octets, each ranging from 0 to 255.
- Example IPv4 address: 192.168.1.1 — four octets separated by dots.
- The value of each octet is calculated through IP addressing and subnetting.
- IP addresses follow protocols — standardised rules that force all devices to communicate in the same language.
- Devices can belong to both private networks (your home/office) and the public network (the Internet).

3.2 IPv4 vs. IPv6

There are two versions of the Internet Protocol in use today. IPv4 has been the standard since the early days of the Internet, but the explosive growth of connected devices created an address shortage that IPv6 was designed to solve.

Feature	IPv4	IPv6
Address Format	Four octets of decimals (e.g. 192.168.1.1)	Eight groups of hex (e.g. 2001:0db8:85a3::8a2e:0370:7334)
Address Space	2^{32} = ~4.3 billion addresses	2^{128} = 340 trillion+ addresses
Exhaustion	Nearly exhausted — primary reason IPv6 needed	Essentially unlimited for the foreseeable future
Notation	Dotted decimal	Colon-separated hexadecimal
Current Status	Still widely used — most of today's Internet	Growing adoption; runs alongside IPv4 (dual-stack)
Security	Security was not a design priority	IPSec encryption built in as a core feature

Why IPv6 Matters for Security

IPv6 changes how devices communicate, how packets are routed, and how firewall rules must be written. A network that supports IPv4 only may accidentally expose IPv6 traffic if the OS has IPv6 enabled by default. Security professionals must understand both.

3.3 Private vs. Public IP Addresses

Not all IP addresses are equal. Devices are assigned either private IPs (used within a local network) or public IPs (used on the Internet). This distinction is fundamental to understanding how home and office networks work.

	Private IP	Public IP
Where Used	Inside a home or office network (LAN)	On the open Internet
Who Assigns	Your router, via DHCP	Your Internet Service Provider (ISP)
Uniqueness	Unique only within the local network	Globally unique across the Internet
Example Range	192.168.x.x / 10.x.x.x / 172.16.x.x	8.8.8.8 (Google DNS), 1.1.1.1 (Cloudflare)
Routable?	Not routable on the public Internet	Fully routable globally

4. MAC Addresses

MAC Address — The Physical Identifier

Every device on a network has a physical network interface — a microchip board embedded in the device's motherboard. This interface is assigned a unique address at the factory where it was manufactured, called a MAC (Media Access Control) address. Unlike IP addresses, MAC addresses are tied to the hardware itself.

- A MAC address is a 12-character hexadecimal number (base-16 numbering system).
- Written in pairs separated by colons — example: a4:c3:f0:85:ac:2d.
- The first six characters (three pairs) identify the manufacturer of the network interface.
- The last six characters (three pairs) are a unique identifier for that specific device.
- MAC addresses operate at Layer 2 (Data Link) of the networking model — handling local delivery.
- IP addresses handle delivery across networks; MAC addresses handle delivery within a single network.

4.1 MAC Address Spoofing

MAC addresses were designed to be permanent hardware identifiers — but they can be faked. The process of changing or impersonating a MAC address is called spoofing, and it has significant security implications.

- Spoofing occurs when a device pretends to have a different MAC address — typically to impersonate another trusted device.
- Security systems that use MAC address filtering (e.g. hotel Wi-Fi, corporate access controls) can be bypassed by spoofing.
- If a firewall allows all traffic from a specific MAC address, an attacker who spoofs that address can bypass the firewall.
- MAC spoofing is also used legitimately — for privacy (randomising MAC on public Wi-Fi) and for network testing.

4.2 Hands-On Lab — MAC Spoofing (Hotel Wi-Fi Simulation)

The TryHackMe room includes an interactive lab simulating a hotel Wi-Fi network. This exercise demonstrates the real-world impact of MAC-based access control and how easily it can be bypassed.

Step	Action	What This Shows
1	**Observe the network**	Alice (green packets) has paid for Wi-Fi and her traffic is allowed through by the router.
2	**Bob's traffic is blocked**	Bob (blue packets) has not paid — the router drops his packets based on MAC address filtering.
3	**Identify Alice's MAC**	Bob looks up Alice's MAC address from the network — it is visible to anyone on the same network segment.
4	**Spoof Alice's MAC address**	Bob changes his MAC address to match Alice's, making the router believe his traffic is coming from her.
5	**Traffic passes through**	The router now forwards Bob's packets, granting access — the MAC-only security is completely bypassed.
6	**Flag captured**	THM{YOU_GOT_ON_TRYHACKME} — demonstrating that MAC filtering alone is not a reliable security control.

Security Lesson

MAC address filtering is a weak security control on its own. It provides a false sense of security because MAC addresses are easily visible on a local network and trivially spoofed. Strong security requires layered controls — authentication, encryption, and monitoring — not just MAC filtering.

5. Ping and ICMP

Ping — The Network's Most Fundamental Tool

Ping is one of the most fundamental and widely used network diagnostic tools. It uses ICMP (Internet Control Message Protocol) packets to test whether a connection between two devices exists and to measure how long data takes to travel between them.

- Ping sends an ICMP Echo Request packet to the target device.
- The target device replies with an ICMP Echo Reply packet.
- The round-trip time (RTT) — measured in milliseconds — tells you how fast the connection is.
- If no reply is received, the device may be offline, unreachable, or blocking ICMP.
- Ping comes pre-installed on Linux, Windows, and macOS — no setup required.
- Pings can be performed against devices on a local network or against websites and remote servers.

5.1 ICMP — Internet Control Message Protocol

ICMP is the protocol that ping uses under the hood. It is a core protocol of the Internet Protocol suite used by networking devices to communicate errors and diagnostic information — not to carry user data.

- ICMP operates at the Network Layer (Layer 3) of the networking model.
- Echo Request (Type 8) and Echo Reply (Type 0) are the specific ICMP message types used by ping.
- Other ICMP messages include: Destination Unreachable, Time Exceeded (used by traceroute), and Redirect.
- Many firewalls block ICMP traffic — this is why some hosts don't respond to ping even when they are online.

5.2 Ping Syntax & Usage

The syntax for ping is straightforward and consistent across operating systems:

```
# Ping an IP address
ping 192.168.1.254

# Ping a website by domain name
ping google.com

# Ping Google's public DNS server (used in the THM lab)
ping 8.8.8.8
```

```
# Ping with a specific number of packets (Windows: -n, Linux: -c)
ping -c 4 8.8.8.8
```

5.3 Reading Ping Output

Understanding what ping tells you is as important as knowing how to run it. Here is what a typical ping output contains:

Output Field	What It Means
bytes=32	Size of the ICMP packet sent (32 bytes is the default on Windows).
time=4ms	Round-trip time — how long the packet took to reach the target and return. Lower is better.
TTL=64	Time To Live — the number of network hops remaining before the packet is discarded. Helps identify the OS of the target.
Packets Sent	Total number of ICMP Echo Requests sent during the session.
Packets Received	Total replies received. If less than sent, some packets were lost.
Packet Loss %	The percentage of packets that did not get a reply. High loss = poor or unreliable connection.
Min/Avg/Max RTT	Summary statistics for round-trip times across all packets sent.

5.4 Hands-On Lab — Pinging 8.8.8.8

The TryHackMe room ends with a practical ping exercise using the deployed interactive website. Learners ping Google's public DNS server (8.8.8.8) to confirm the connection exists and retrieve the flag.

```
$ ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=4ms TTL=118
Reply from 8.8.8.8: bytes=32 time=3ms TTL=118
Reply from 8.8.8.8: bytes=32 time=5ms TTL=118
Reply from 8.8.8.8: bytes=32 time=4ms TTL=118

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 5ms, Average = 4ms
```

Flag Captured

After completing the ping exercise against 8.8.8.8, the TryHackMe lab reveals the flag confirming successful completion. This demonstrates that ping is not just a diagnostic tool — it is the first command a network professional runs to verify connectivity.

6. Core Concepts Summary

#	Concept	Definition
1	**Network**	A group of devices connected together so they can communicate.
2	**Internet**	A global network of smaller networks — invented by Tim Berners-Lee (WWW in 1989).
3	**IP Address**	A logical identifier for a device on a network, made of four octets (e.g. 192.168.1.1).
4	**Octet**	Each of the four sections of an IPv4 address — a number from 0 to 255.
5	**IPv4**	Current dominant IP standard — supports ~4.3 billion addresses (2^{32}).
6	**IPv6**	Newer IP standard — supports 340+ trillion addresses (2^{128}), with built-in IPSec.
7	**Private IP**	IP address used within a local network (e.g. 192.168.x.x). Not routable on the Internet.
8	**Public IP**	Globally unique IP address assigned by your ISP. Routable on the Internet.
9	**MAC Address**	A 12-character hexadecimal hardware identifier assigned to a device's network interface at manufacture.
10	**MAC Spoofing**	The process of faking a MAC address to impersonate another device on the network.
11	**Ping**	A network diagnostic tool that uses ICMP to test connectivity and measure latency.
12	**ICMP**	Internet Control Message Protocol — carries ping (Echo Request/Reply) and network error messages.
13	**RTT**	Round-Trip Time — the time for a packet to travel to a target and back, measured in milliseconds.
14	**TTL**	Time To Live — a counter limiting how many hops a packet can travel before being discarded.

7. Security Implications — Why This Matters

IP Addresses Are Not Permanent Identifiers

Because IP addresses can be reassigned (and spoofed), they cannot be trusted as definitive proof of identity. Attackers routinely use IP spoofing, VPNs, proxies, and Tor to mask their real source IP. Security tools must combine IP data with other signals for reliable attribution.

MAC Filtering Is a Weak Control

The hotel Wi-Fi lab demonstrated that MAC-based access control is trivially bypassed by spoofing. Any security control that relies solely on MAC addresses for authentication provides only the illusion of security. Use it as a supplementary layer only, never as a primary control.

Ping Is Both a Tool and a Recon Method

Penetration testers and attackers use ping (and ICMP probing broadly) during the reconnaissance phase to discover live hosts on a network. A host that responds to ping confirms it is online and reachable. This is why many organisations configure firewalls to block ICMP — but this can also interfere with legitimate diagnostics.

IPv6 Introduces New Attack Surface

Many networks enable IPv6 by default on operating systems without properly configuring firewall rules for it. This creates a blind spot — security tools tuned for IPv4 may miss IPv6 traffic entirely. Attackers can use IPv6 to bypass IPv4-focused security controls.

Every Attack Starts with Network Understanding

Reconnaissance — the first phase of any attack — is fundamentally a networking exercise. Identifying live hosts (ping), open ports (Nmap), services, and network topology all require a solid understanding of IP addresses, protocols, and how devices communicate. Networking knowledge is not optional for security work.

8. Key Takeaways & Next Steps

8.1 Key Takeaways

- Networks are collections of connected devices — from two computers to the global Internet.
- The Internet is a network of networks; Tim Berners-Lee invented the World Wide Web in 1989.
- Every device has two identifiers: an IP address (logical, can change) and a MAC address (physical, hardware-assigned).

- IPv4 uses four octets (e.g. 192.168.1.1); IPv6 uses 128-bit hex addresses to solve the exhaustion problem.
- Private IPs (192.168.x.x, 10.x.x.x) are used inside local networks; public IPs are used on the Internet.
- MAC addresses are 12-character hex values where the first 6 identify the manufacturer.
- MAC spoofing can bypass MAC-based access controls — not a reliable security mechanism alone.
- Ping uses ICMP Echo Request/Reply to test connectivity and measure round-trip time.
- TTL in ping output indicates how many hops remain — and can reveal the target operating system.

8.2 Next Steps on TryHackMe

- **Immediate:** Continue with the Intro to LAN room — covers LAN topologies, subnetting, and ARP.
- **Foundation:** Complete the OSI Model room — understand the 7-layer model that governs all networking.
- **Foundation:** Work through the Packets & Frames room — how data is broken up and reassembled.
- **Essential:** Study the DNS in Detail room — how domain names resolve to IP addresses.
- **Essential:** Take the HTTP in Detail room — how web traffic works, critical for web security.
- **Path:** Complete the Pre-Security learning path in full before moving to offensive or defensive paths.

Why Start with Networking?

Every room, every exploit, every defence tool you encounter in cyber security will require you to understand how devices communicate over a network. Time invested in networking fundamentals pays dividends across your entire career — both offensive and defensive. Get comfortable here before moving on.

Source: TryHackMe — What is Networking? Room | tryhackme.com/room/whatisnetworking